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Scoped USDC payments for AI agents



2026 FinTechathon / Topic F

Mandate Labs

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Spending authority for AI agents

The owner defines the scope

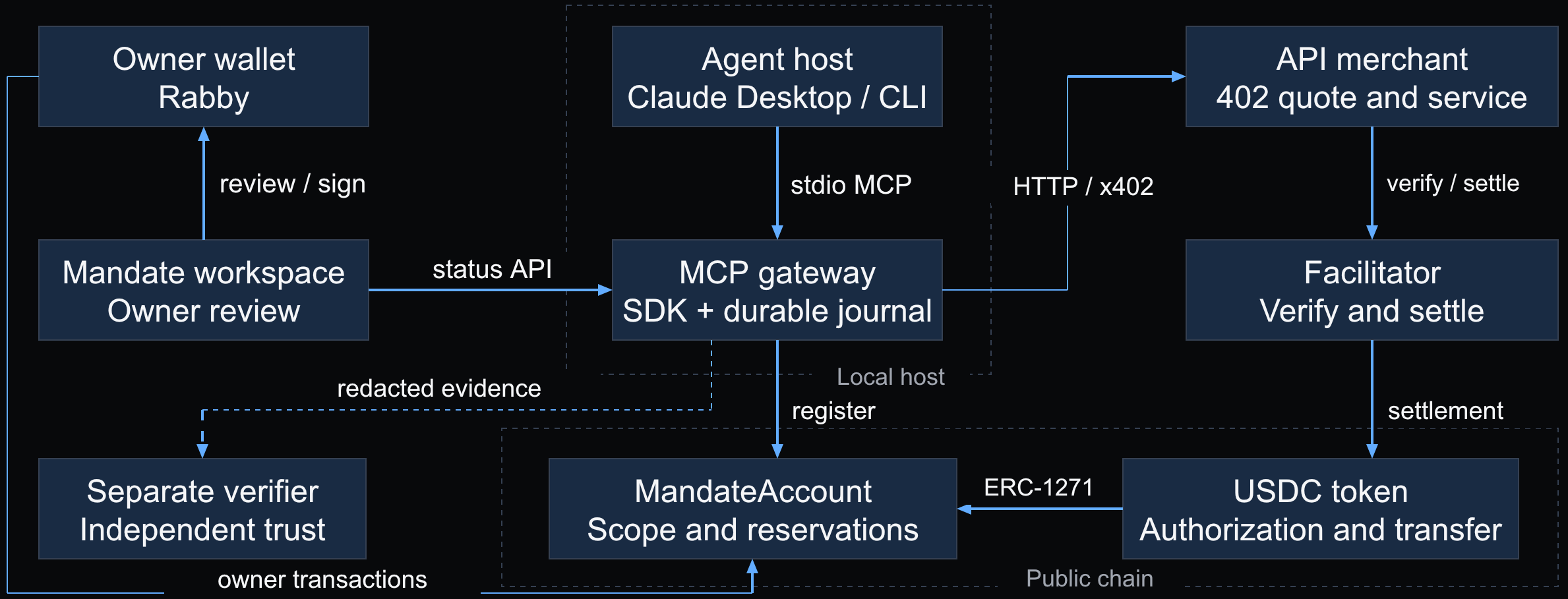
Approved service, payee, asset, budget and expiry.

The agent executes the task

A finite session authorizes specific purchases.

Contract checks continue when a host uses its session key directly.

System architecture and trust boundaries



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A bounded ETH data purchase

0.15

total test USDC budget

0.10

maximum per payment

0.01

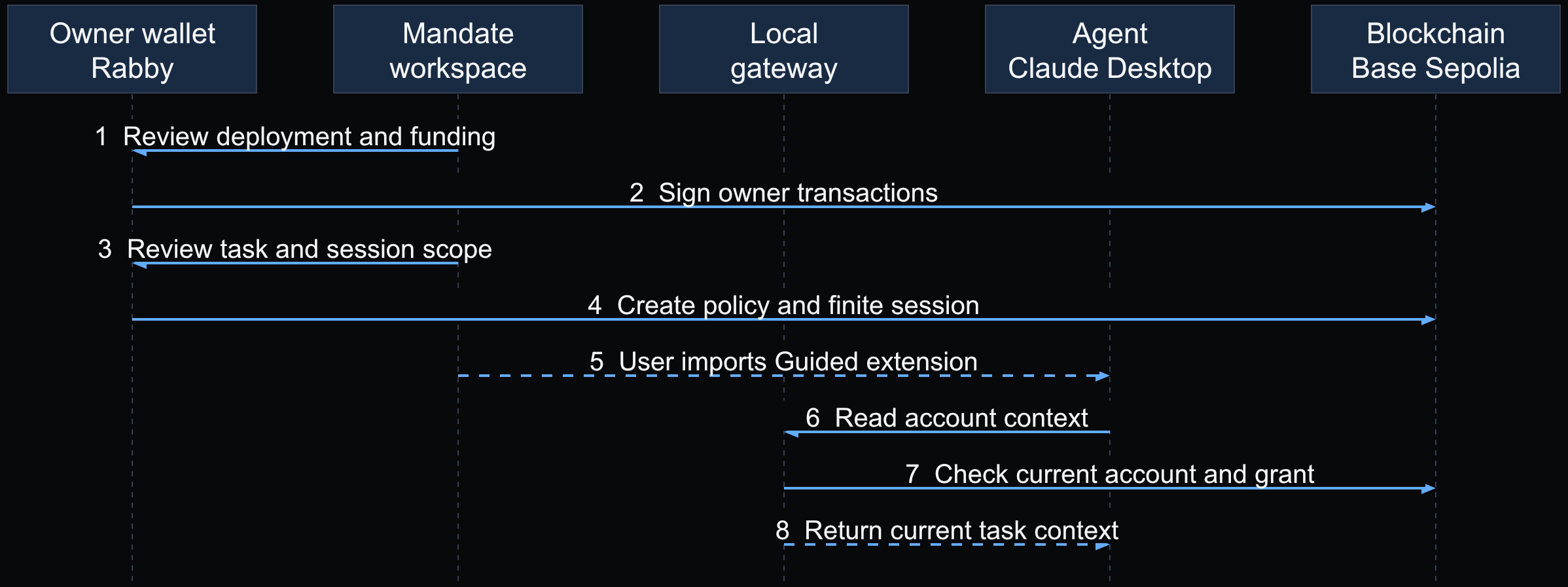
price of the ETH sample

Claude Desktop + Rabby + Base Sepolia

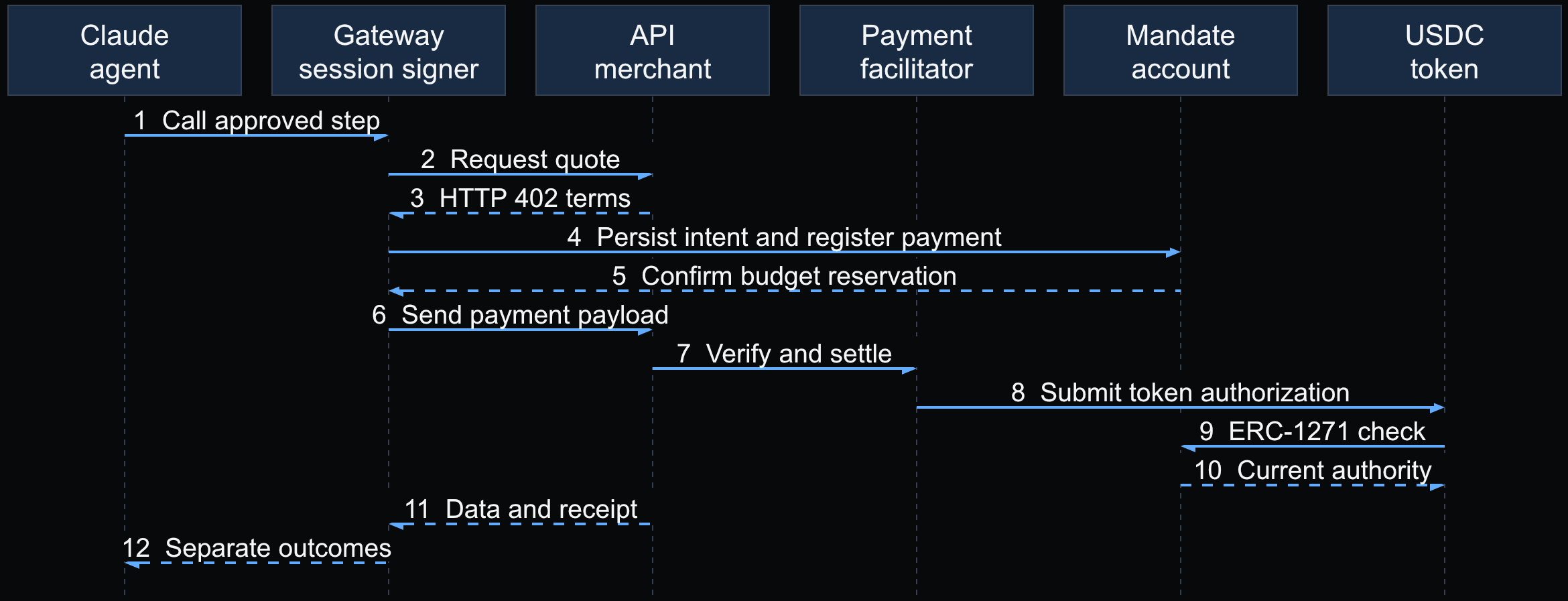
Fixed ETH dataset. Test USDC payment. Gas uses separate test ETH.

Recorded session: 2 hours. Each payment authorization: at most 5 minutes.

Account preparation and Claude connection



x402 payment registration and settlement



Shared budgets and payment identity

occupied(root) ≤ budget(root) occupied(session) ≤ budget(session)

Constraint	Implemented rule
Registration	EIP-712 binds chain, wallet, policy, session and payment
Shared budget	Atomic root and session reservations
Business retry	Permanent used step and deterministic nonce
Settlement	Registered digest with live version and epoch
Safe release	Strict expiry and unused token nonce

Concurrent 0.10 requests against 0.15: at most one reserves before release.

Recovery preserves the original payment

Observation	Payment	Delivery	Result
First response: HTTP 500	paid	failed	unchecked
Original-order recovery	paid	returned	valid

One 0.01 USDC settlement

Same order, business nonce and transaction hash throughout recovery.

Research questions and implemented mechanisms

01 **Policy-bound actions**

Typed registration and shared reservations

02 **Delegated-session evidence**

Signed checkpoints and selective disclosure

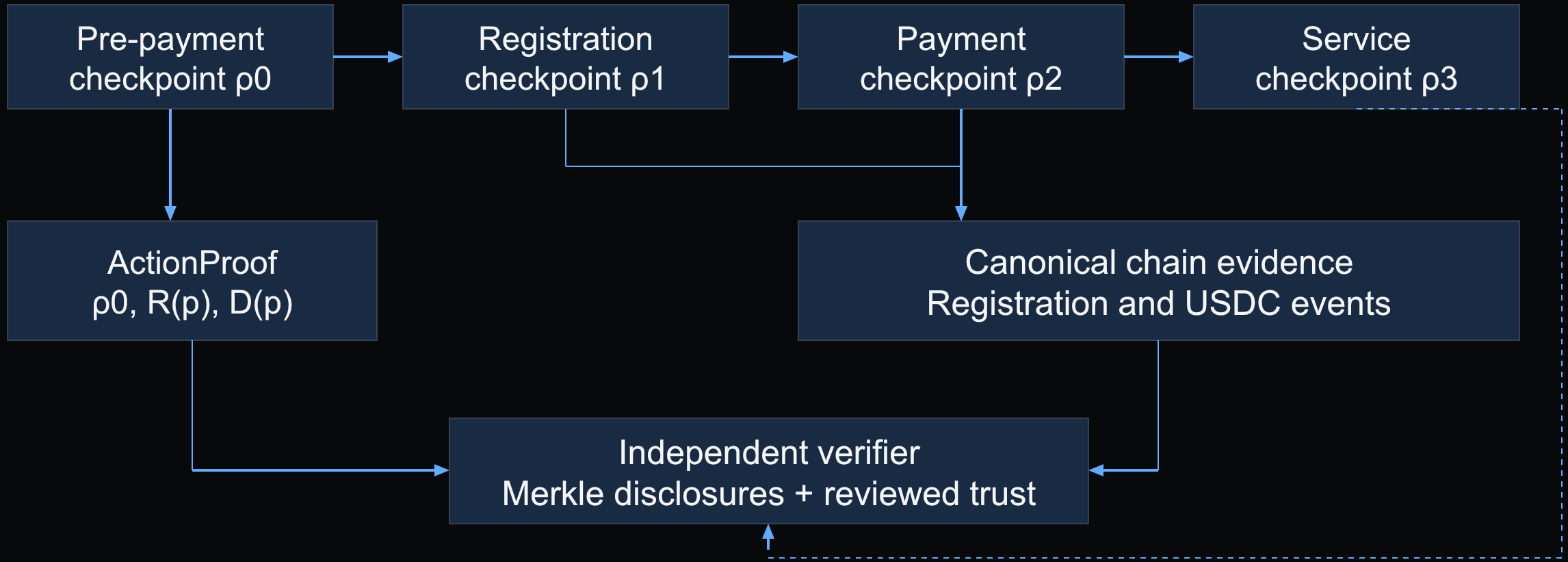
03 **Host interference**

Reference controller and retained-head checks

The release implements a bounded part of the prior proposal.

Session evidence and independent verification

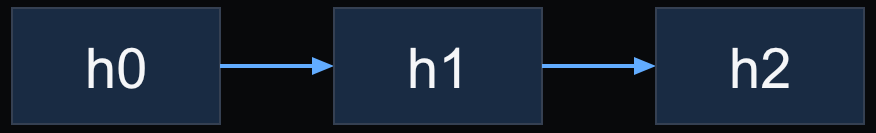
Signed headers bind scope, field root μ and the previous checkpoint digest.



Required financial checks: passed. Overall: incomplete.

Host interference and missing history

World A: history ends at checkpoint 2



World B: checkpoint 3 is withheld



Same disclosed prefix

No independent latest head
Completeness remains unknown

An externally retained head can expose a missing known tail.

A query cap applies at the controller interface. The host can bypass that interface.

Evaluation and measured overhead

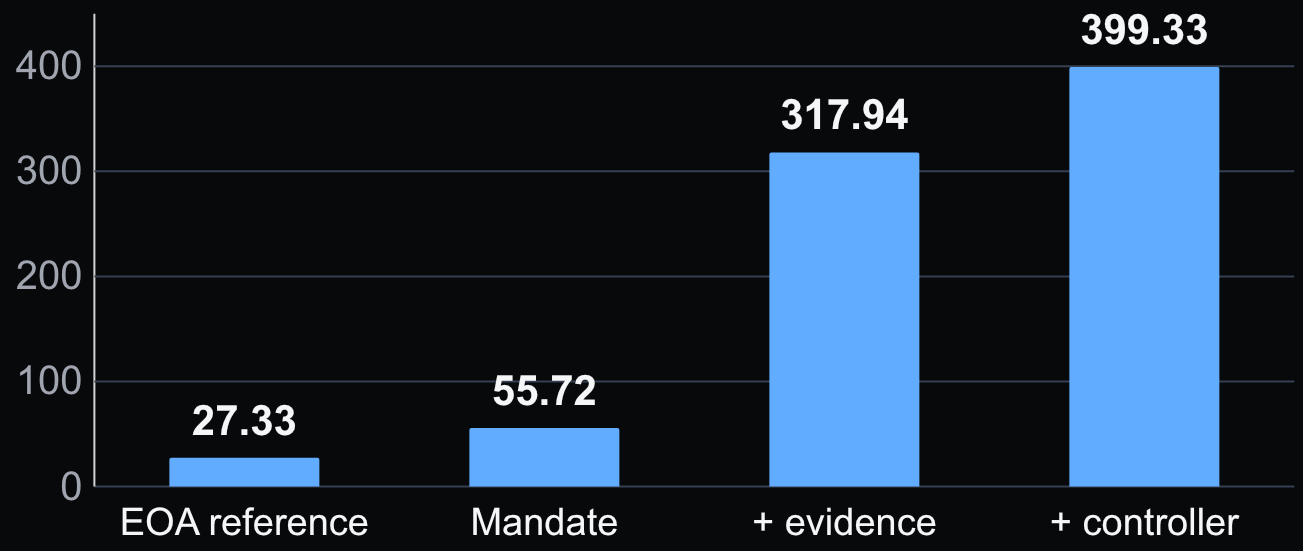
120 tasks

132 settlements
0 duplicate payments

10,000 invalid calls

20 explicit categories
6 legal controls

Local full-task median (ms)



Local chain, n=30 per group. Setup excluded. Applicable export and verification included.

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Public execution and revocation evidence

0.01

test USDC paid once

0.14

final test USDC balance

0

new transactions after refusal

Settlement on BaseScan

Session revocation on BaseScan

Eight exported key transactions. Required financial checks passed.

Controlled merchant and facilitator. HTTPS source and full history remain unproven.

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Financial scope with reviewable evidence

Next implementation priorities

Supported installation and updates

Broader merchant compatibility

Independent source and history evidence

Beacon commitments and stronger execution proofs remain research extensions.

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